

Statistical Arbitrage in Cryptocurrencies

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1. Introduction

This project studies whether recent cryptocurrency price movement contains a useful momentum or reversal signal. It uses BTC/USD as the cleanest long-history time-series test, then extends the same momentum idea to a 17-coin cross-sectional dollar-neutral portfolio. The final submission therefore addresses both a high-quality BTC trend-following test and broader crypto statistical-arbitrage research. The goal is not to prove a permanent trading edge, but to test a disciplined hypothesis with transaction costs and held-out performance.

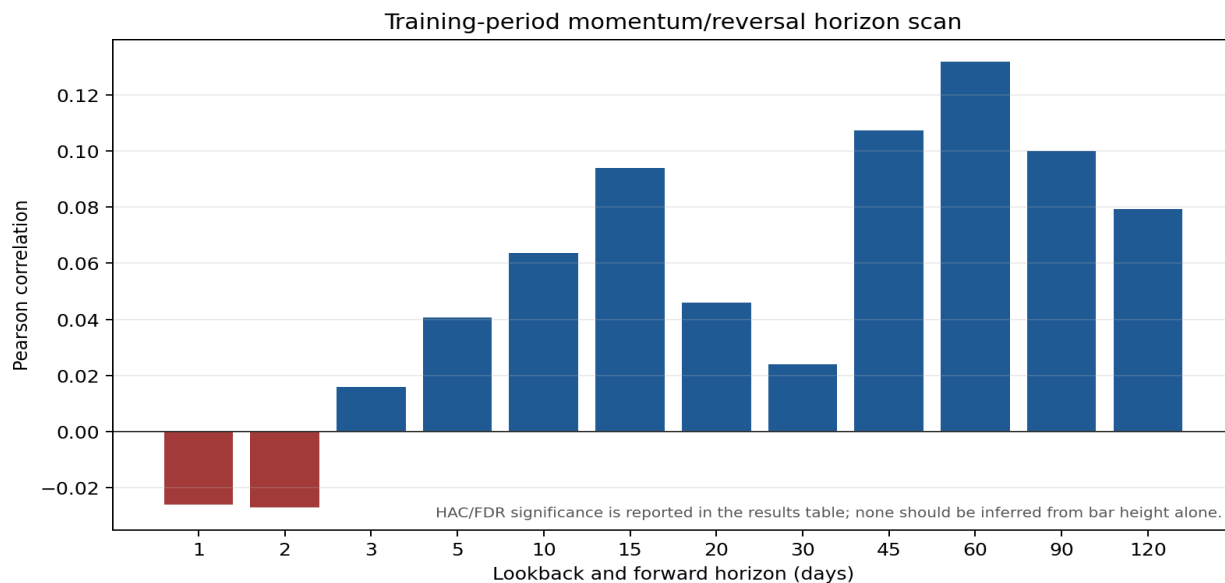
2. Data Collection and Preprocessing

The data is Bitstamp BTC/USD one-minute OHLCV history from 2012-01-01 through 2025-01-07. The minute records are validated, duplicate timestamps and missing values are rejected, incomplete calendar days are removed, and the clean data is converted into complete daily open, high, low, close, and volume observations. The formal analysis begins on 2015-01-10 to avoid the earliest, least-liquid BTC regime.

For the cross-sectional extension, the notebook downloads daily Yahoo Finance crypto data for this universe: BTC-USD, ETH-USD, BNB-USD, XRP-USD, ADA-USD, DOGE-USD, MATIC-USD, LTC-USD, BCH-USD, LINK-USD, XLM-USD, ATOM-USD, ETC-USD, FIL-USD, TRX-USD, EOS-USD, and XMR-USD.

3. Momentum and Reversal Research

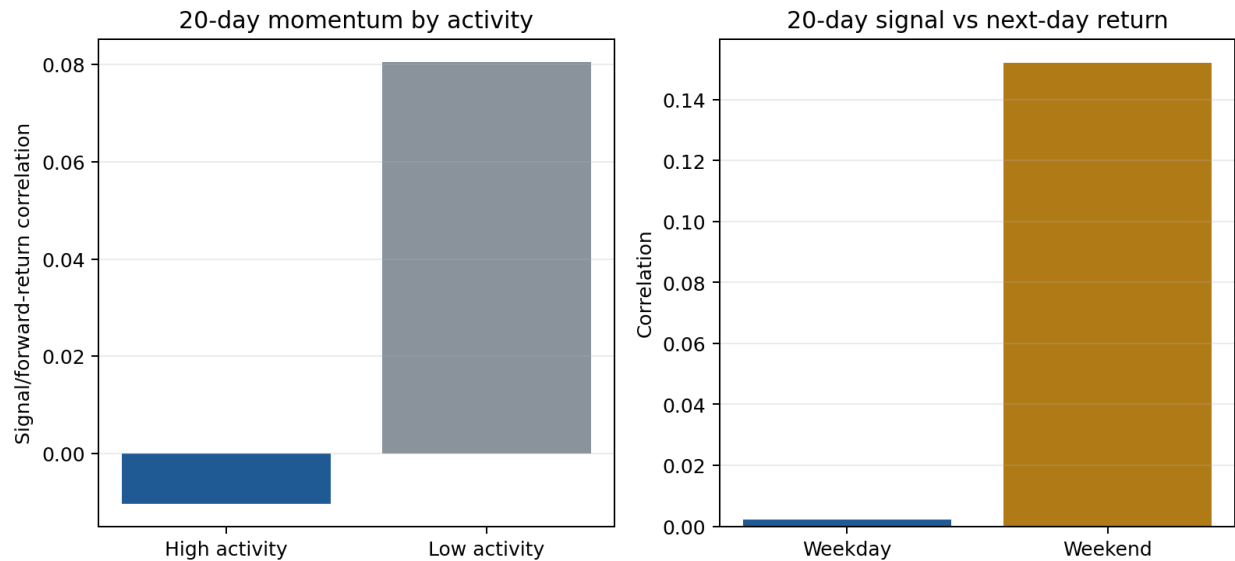
The main research test compares past log returns with future log returns across 1, 2, 3, 5, 10, 15, 20, 30, 45, 60, 90, and 120 day horizons. Positive correlations suggest momentum; negative correlations suggest reversal. Because overlapping multi-day returns are not independent, the project reports HAC-adjusted inference and false-discovery-rate adjustments.



Training-period results were suggestive but not definitive: medium and long horizons were directionally positive, while no horizon remained significant at the 5% level after the final statistical corrections.

4. Activity and Seasonality Tests

The project also asks whether the signal behaves differently during higher-volume periods or when trades are entered on weekends versus weekdays. These tests were included because crypto trades continuously and market participation can vary by time and activity level. The evidence did not strongly support using either feature as the final trading filter.



5. Strategy Selection and Backtest Design

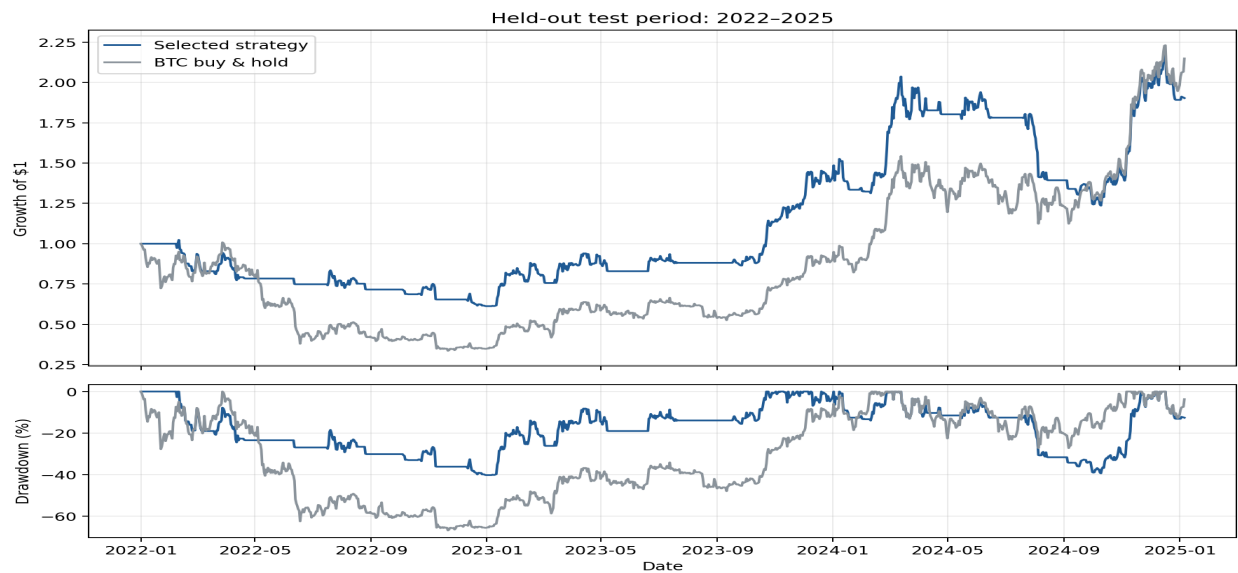
The sample is split into training, validation, and held-out test periods. Candidate rules vary the momentum horizon, long-only versus long/short positioning, and the activity filter. The selected rule is chosen using validation Sharpe ratio only, with lower turnover used as a tie-breaker. The final test period is not used for selection.

Period	Dates	Purpose
Training	2015-01-10 to 2019-12-31	Research signal family
Validation	2020-01-01 to 2021-12-31	Choose final configuration
Test	2022-01-01 to 2025-01-06	Evaluate selected rule

The selected strategy uses a 30-day lookahead, long-only positioning, no activity filter, and a 20 bps market-order cost. If the trailing return is positive, the strategy holds Bitcoin the next day; otherwise it remains flat. The one-day shift prevents look-ahead bias.

6. Performance

Metric	Selected strategy	BTC buy and hold
Total return	90.3%	119.3%
CAGR	23.8%	29.7%
Annualized volatility	36.7%	53.7%
Sharpe ratio	0.76	0.75
Maximum drawdown	-40.2%	-66.8%
Annualized alpha	9.1%	N/A
Beta	0.47	N/A



The selected strategy was profitable during the held-out period and produced lower volatility and a smaller drawdown than holding Bitcoin continuously. It earned less total return than buy-and-hold, but had a slightly higher 365-day Sharpe ratio. Its beta of 0.47, BTC correlation of 0.68, and R-squared of 0.47 show meaningful BTC exposure, so the strategy should be framed as BTC market timing rather than independent alpha.

7. Exposure, Turnover, and Year-by-Year Stability

Diagnostic	Value
HAC alpha t-stat	0.59
HAC alpha p-value	0.554
BTC beta	0.47
BTC correlation	0.68
R-squared	0.47
Sharpe, 252-day convention	0.63
Annualized turnover	32.46
Trade days per year	32.46

Year	Strategy total	Strategy Sharpe	BTC total	BTC Sharpe
2022	-38.7%	-1.44	-64.5%	-1.37
2023	129.0%	2.38	157.4%	2.25
2024	34.9%	0.93	119.0%	1.78
2025	0.6%	3.86	9.5%	19.08

8. Trading Costs and Robustness

Execution costs are economically meaningful: test Sharpe falls from 0.94 before costs to 0.76 under the 20-bps market-order assumption.

Scenario	Cost	Total return	CAGR	Sharpe	Max drawdown
zero_cost	0 bps	131.6%	32.1%	0.94	-36.6%
limit_order_7bps	7 bps	116.2%	29.1%	0.88	-37.6%
market_order_20bps	20 bps	90.3%	23.8%	0.76	-40.2%

9. Cross-Sectional Crypto Extension

To address the main limitation of trading only BTC, the project adds a 17-coin dollar-neutral extension. It ranks coins by 14-day momentum, rebalances every 14 days, goes long the top 3 coins and short the bottom 3, with 1.0 gross and 0.0 net exposure.

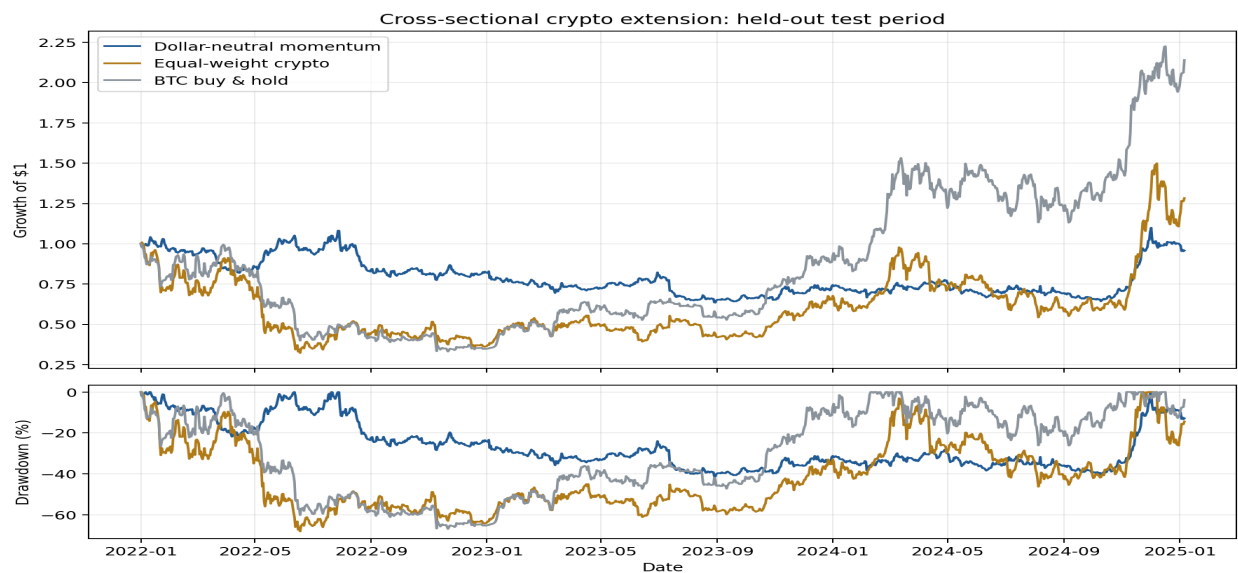
The tested universe is BTC, ETH, BNB, XRP, ADA, DOGE, MATIC, LTC, BCH, LINK, XLM, ATOM, ETC, FIL, TRX, EOS, and XMR, all represented as USD pairs from Yahoo Finance daily price data.

Metric	Dollar-neutral momentum	Equal-weight crypto	BTC buy and hold
Total return	-2.1%	33.4%	120.4%
CAGR	-0.7%	10.0%	29.9%
Sharpe ratio	0.10	0.47	0.75
Maximum drawdown	-41.4%	-68.1%	-66.9%
BTC beta	-0.00	0.98	N/A
BTC correlation	-0.01	0.83	N/A

Direct Coin-by-Coin Replication

The exact 30-day long/flat BTC rule was also applied independently to all 17 coins with 20 bps costs. In the held-out test, 12/17 coins had positive returns, 13/17 had positive Sharpe ratios, and 8/17 beat buy-and-hold on Sharpe. The median strategy return was 20.0% versus 0.2% for buy-and-hold. The result is broad but uneven, so it is evidence of a risk-management pattern rather than proof of a universal crypto edge.

Coin	Rule total	Buy-hold total	Rule Sharpe	Buy-hold Sharpe
DOGE	228.6%	127.5%	0.88	0.75
BTC	95.7%	120.4%	0.78	0.75
XLM	105.7%	66.7%	0.64	0.59
EOS	77.8%	-69.8%	0.61	-0.06
XRP	43.3%	191.0%	0.48	0.80
ADA	40.4%	-16.5%	0.47	0.33
ATOM	29.5%	-77.0%	0.45	-0.11
FIL	17.8%	-82.5%	0.43	-0.07
TRX	27.6%	256.5%	0.39	0.87
ETC	14.5%	-16.0%	0.37	0.37
BNB	20.0%	42.5%	0.35	0.50
BCH	-2.0%	12.1%	0.30	0.46
ETH	7.6%	0.2%	0.28	0.34
MATIC	-48.6%	-79.3%	-0.07	-0.10
LINK	-63.7%	21.6%	-0.21	0.51
LTC	-72.7%	-22.2%	-0.54	0.27
XMR	-82.0%	-10.9%	-0.87	0.31



The extension reduces BTC dependence almost completely, but the after-cost test return is -2.1% and Sharpe is 0.10. This means the stronger conclusion is that the BTC timing rule improves risk, while the broader market-neutral momentum effect is not robust in this test.

10. Statistical Significance

The strategy's daily returns are regressed against BTC daily returns using HAC standard errors. The test-period alpha estimate is positive, but its t-statistic is only 0.59, so it is not statistically significant at the final threshold. This keeps the conclusion appropriately conservative: the backtest is useful evidence for a defensive momentum rule, not proof of a guaranteed edge.

11. Conclusion

The final project demonstrates a reproducible cryptocurrency momentum study. A 30-day long-only Bitcoin rule was selected without using the held-out test period, remained profitable after 20 bps trading costs, and reduced volatility and drawdown relative to buy-and-hold. The 17-coin dollar-neutral extension addresses the BTC-only limitation and removes most BTC exposure, but its weak after-cost held-out result means the project should not claim robust independent alpha. The strongest next improvements are exchange-level liquidity filters, funding and borrow-cost data, reversal baskets, and rolling walk-forward validation.